X-MPFD-E++: Response to Tentative Checklist (AR1115)

Changes made to the specification, supporting data and drawings

To	Adv. Pranav Bhat, Patent Counsel, Lextria Research / IPTTO, Manipal Academy of Higher Education
From	The Inventors: Panchadip Bhattacharjee, Somyajeet Arukh, Sanvi Sharma, Saloni Ruia, Gururaj HL
Invention	AR1115: Deterministic, Machine-Executable Multimodal System and Method for Rapid Zero-Day Phishing and Fraud Detection with Conditional Hardware Gating
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1. Summary of responses

Each of the six points in the Tentative Checklist is answered below. For every point we state what was clarified or added, give the proposed wording for the Complete Specification where the point concerns the description of the invention, and identify the file that supports it.

#	Checklist point	Response	Supporting file
1	URL normalisation before entropy	Normalisation steps stated: Unicode NFC, one percent-decoding pass, lower-casing of scheme and authority, default-port removal, path slash handling	orchestrator.py (URLNormalizer)
2	Node types counted by the tag-stack state machine	Counting rule stated for each node type, including void, script, iframe and implicitly closed elements	orchestrator.py (DeterministicDOMAnalyzer)
3	Latency of 3.971 ms (gated) vs 6.621 ms (ungated), and the operations timed for a bypassed input	Per-operation microsecond breakdown measured; the two measurement bases are explained	dga_latency_breakdown_results.json; FIG. 5
4	Late- and early-fusion reference values and their basis	Weights, decision boundaries and basis of selection stated	fusion_engine.py (Tables 6-1, 6-2)
5	Benchmark, hold-out, entropy profile, tests and examples	Result files and execution logs compiled; hold-out run at three entropy scopes	See Section 6
6	Graphs and plots under Section 10(4)	Seven drawings supplied: two new block diagrams and five data plots	figures/ (FIG. ARCH, FIG. FLOW, FIG. 1 to FIG. 5)

2. Item 1: URL normalisation before the entropy is computed

Before character frequencies are counted for the Shannon entropy $H(U) = -\Sigma\ P(x_i)\ \log_2 P(x_i)$, the URL is processed by the following steps, in this order.

Step	Operation	Detail
1	Unicode normalisation	Form NFC (canonical decomposition followed by canonical composition, Unicode Standard Annex #15), applied after stripping surrounding whitespace.
2	Percent-decoding	Decoded exactly once. A second pass is never made, so a doubly encoded sequence is not decoded twice.
3	Case conversion	Scheme and authority (host) converted to lower case, following RFC 3986 sections 3.1 and 3.2.2. Path, query and fragment keep their original case.
4	Default ports	Port :80 removed for http and :443 removed for https.
5	Path	Consecutive slashes in the path are collapsed to a single slash. An empty path on a non-empty host is set to /.

Step	Operation	Detail
6	Reconstruction	The URL is reassembled from the normalised components. If no scheme is present, http is assumed.

Proposed text for the Complete Specification (Detailed Description):

> *"Prior to tabulating character occurrence frequencies for the Shannon entropy calculation, the input URL string is converted to Unicode Normalization Form NFC and subjected to a single-pass percent-decoding operation. The scheme and the authority component are converted to lower case, while the path, query and fragment retain their case. Default port designations (:80 for http and :443 for https) are removed, consecutive slashes in the path are collapsed to a single slash, and an empty path is replaced by a single slash. The character frequencies, and hence the entropy, are thereby unaffected by letter-case variation in the host and by a single layer of percent-encoding."*

3. Item 2: node types counted by the tag-stack state machine

The analyser is a tag-stack state machine that counts element nodes only. N_{total} is the number of element nodes and D_{max} is the greatest stack depth reached.

Node type	Counted in N_{total}	Updates D_{max}	Pushed on stack	Note
Element (non-void)	Yes	Yes	Yes	Popped when its end tag, or an ancestor's end tag, is met
Void element (area, base, br, col, embed, hr, img, input, link, meta, param, source, track, wbr)	Yes	Yes, as a leaf	No	Void elements have no end tag; pushing them would raise the depth of later siblings
Text	No	No	No	Excluded so that SDR does not depend on the amount of prose
Comment	No	No	No	Does not render and does not change the hierarchy
Script	Yes, as one node	Yes	Yes	Script source is treated as data and is not parsed into nodes
Iframe	Yes, as one node	Yes	Yes	The document referenced by the iframe is not fetched or parsed
Implicitly closed element	Yes	Yes	Yes	Unwound when an ancestor end tag is met or at end of input

The Structural Density Ratio and its deviation from a baseline are defined as follows.

$$SDR = N_{total} / \max(1, D_{max}) \quad \Delta SDR = |SDR - SDR_{baseline}| / SDR_{baseline} \times 100\%$$

A cloned credential harvester is flagged when ΔSDR exceeds 50%, D_{max} is 4 or less, and the page contains an input of type password. In the measured corpus, authentic portals have D_{max} of 12 or more and SDR of 13.0 or more, while the cloned harvesters have D_{max} of 4 or less and SDR of 3.5 or less (FIG. 3).

Proposed text for the Complete Specification (Detailed Description):

> *"The tag-stack state machine counts only element nodes when determining the total node count and the maximum tree depth. Void elements are counted as leaf nodes and are not pushed onto the stack. Text nodes and comment nodes are excluded. A script element is counted as a single node and its source is not parsed into nodes. An iframe element is counted as a single node and the document it references is neither fetched nor parsed. Elements that are not explicitly closed are removed from the stack when an ancestor end tag is encountered or at the end of the input."*

4. Item 3: latency on the domain-generated subset

The two pairs of figures are measured on different bases. Both are reported so that the effect can be read at the level of the operations and at the level of a complete run.

Basis A: isolated per-operation profile. A profiler (dga_latency_breakdown.py) times each operation with a nanosecond counter, over 200 iterations for each of the test patterns (2,000 measurements). Mean values are shown below.

Operation	Gated (AST bypassed)	Ungated (baseline)
1. URL normalisation (NFC, single unquote, lower-casing, port removal)	36.6 µs	36.6 µs
2. Lexical Shannon entropy H(U)	42.8 µs	42.8 µs
3. Gate comparison and branch	0.3 µs	0.3 µs (shown in both bars in FIG. 5)
4. Feature registry and SHA-256 provenance hash	27.3 µs	27.3 µs
5. Fusion rule evaluation (Table 6-1, Rule 1; withhold directive)	17.5 µs	17.5 µs
6. Explanation synthesis and audit hash	40.9 µs	40.9 µs
7. HTML AST parsing, tag-stack maintenance and SDR	bypassed	646.3 µs
Total per input	**165.4 µs**	**811.7 µs**

Operation 7 is the only difference between the two pipelines. Withholding it saves 646.3 µs per input, a reduction of 79.6% (FIG. 5). The profiler file records the comparator as run only in the gated pipeline; FIG. 5 draws it in both bars so that the first six segments can be compared like for like (difference 0.3 µs).

Basis B: complete benchmark run. The figures 3.971 ms (gated) and 6.621 ms (ungated) are mean times per input over the complete 1,000-input benchmark, a difference of 2.650 ms (40.0%). They are not the time of the DGA classification alone. Each input is timed together with the work of the benchmark harness around it: socket handling, formatting of the report as canonical JSON (RFC 8785), writing of the audit log to disk, and garbage collection. The means also include the 333 authentic enterprise portals, which are fully parsed and take 4.0 to 10.0 ms each. The distribution of times over all inputs is skewed towards long times, so the mean is higher than the median of 1.26 ms (FIG. 1). For the 334 bypassed DGA inputs taken alone, the gated time per input has a median of 0.231 ms (minimum 0.192 ms, maximum 1.226 ms), which agrees with the profiled total of Basis A.

Proposed text for the Complete Specification (Detailed Description):

> *"When the entropy of the normalised URL exceeds the threshold, the system performs URL normalisation, entropy calculation, the threshold comparison, feature registration with a SHA-256 provenance hash, evaluation of the fusion rule, and synthesis of the explanation and audit record. HTML parsing, maintenance of the tag stack and computation of the structural density ratio are not performed. In measurement, omission of these operations reduced the processing time of a domain-generated input by 646.3 microseconds, which is 79.6% of the time taken when they are performed. The processing time of a bypassed input had a median of 0.231 milliseconds."*

5. Item 4: fusion reference values and basis of selection

Late confidence-weighted fusion (Claim 5). Composite risk $R_{late} = w_L \cdot S_L + w_S \cdot S_S + w_M \cdot S_M$, where S_L = clip((H(U) − 2.50) / 2.50, 0, 1), S_S = clip(ΔSDR / 100, 0, 1) when HTML is available (otherwise 0), and S_M = 0.95 if the TLS certificate is invalid or self-signed (otherwise 0.05).

Quantity	Value
Weights [w_L, w_S, w_M]	[0.40, 0.40, 0.20]. If HTML is absent: [0.667, 0, 0.333]
Decision boundaries	$R_{late} \geq 0.65$: phishing. $0.40 \leq R_{late} < 0.65$: suspicious. $R_{late} < 0.40$: legitimate
Basis of selection	The threshold 0.65 was selected on the 1,000-input comparative benchmark corpus (334 domain-generated, 333 cloned, 333 legitimate) as the value that maximises Youden's index J = sensitivity + specificity − 1 (J = 0.994)

Early fusion (Claims 11 and 12). The signals are projected into a vector $X = [x_1, x_2, x_3]^T$, with x_1 = clip(H(U) / 6.0, 0, 1), x_2 = clip(ΔSDR / 100, 0, 1) and x_3 = 0 if the TLS certificate is valid, otherwise 1. The scalar risk is $R_{early} = W^T X$.

Quantity	Value
Projection weights W	$[0.45, 0.35, 0.20]^T$
Decision boundaries	$R_{early} \geq 0.60$: phishing. $0.35 \leq R_{early} < 0.60$: suspicious. $R_{early} < 0.35$: legitimate
Basis of selection	Lexical entropy has the largest weight (0.45) because it gives the earliest signal for generated domains; structural deviation has 0.35; the transport signal has 0.20 as corroboration. The weights and the boundaries were set by engineering judgement and were not fitted to data. They were then checked on the 1,000-input comparative corpus, where early fusion gave an area under the ROC curve of 0.995 (FIG. 4). The boundary 0.60 is placed so that a maximal lexical signal together with a maximal structural signal (0.45 + 0.35 = 0.80), or together with an invalid TLS certificate (0.45 + 0.20 = 0.65), is classed as phishing. The boundary 0.35 was set in the same way.

Gated fusion (Claims 6 and 15). If H(U) exceeds the gating threshold τ, the system returns PHISHING_DGA with risk 0.99. Otherwise the HTML is parsed. If ΔSDR exceeds 50% and the TLS certificate is invalid, the result is PHISHING_CLONE (risk 0.95). If ΔSDR exceeds 50% and the certificate is valid, the result is SUSPICIOUS (risk 0.70). In all other cases the result is LEGITIMATE (risk 0.05). The threshold τ is configurable for three entropy scopes: 4.50 bits/char over the full URL, 4.15 over host and path, and 3.80 over the hostname (Section 6).

6. Item 5: benchmark, hold-out, entropy profile, tests and examples

File	Content
comparative_benchmark_results.json	1,000 inputs (334 DGA, 333 cloned harvesters, 333 legitimate). Gated: mean 3.833 ms, median 1.263 ms. Ungated: mean 5.478 ms, median 6.606 ms. 334 AST parses bypassed. For the bypassed DGA inputs alone, gated median 0.231 ms (range 0.192 to 1.226 ms). Accuracy, precision and recall 100%, false positive rate 0%
ANNEXURE_B_INDEPENDENT_HOLDOUT_LOGS.json	500 unseen inputs (200 legitimate, 160 DGA, 140 cloned harvesters), run at the three entropy scopes (table below)
test_fresh_corpus.py	Second independent 500-input corpus, run at the same three scopes (table below)
legitimate_entropy_profile.json	90 authentic enterprise portals (the sample shown in FIG. 2). Mean H(U) 3.87, 95th percentile 4.21, maximum 4.34 bits/char. The lowest value among the DGA domains is 4.51, a margin of 0.174 bits/char
dga_latency_breakdown_results.json	Per-operation timings for Item 3 (2,000 measurements)
test_execution_log.txt	24 automated unit tests, all passed (0.024 s)
examples_execution_log.txt	9 worked end-to-end examples, all executed

Hold-out results by entropy scope. Percentages in brackets are of the 500 inputs (bypasses) or of the legitimate inputs in the corpus (false positives).

Metric	Full URL (τ = 4.50)	Host + path (τ = 4.15)	Hostname (τ = 3.80)
AST parses bypassed	191 (38.2%)	262 (52.4%)	291 (58.2%)
Median latency	1.41 ms	1.18 ms	0.79 ms
False positives, corpus 1	22 of 200 (11.0%)	0 of 200 (0.00%)	0 of 200 (0.00%)
False positives, corpus 2	37.0%	5.0%	0.00%
Threat recall	300 of 300 (100%)	300 of 300 (100%)	300 of 300 (100%)

The false positives at the full-URL scope arise from long hexadecimal session tokens in query strings, which raise the entropy of the whole URL. Computing the entropy over the host and path, or the hostname alone, removes the effect of the query string. This is the reason the scope is configurable.

7. Item 6: drawings under Section 10(4)

Seven drawings are supplied. FIG. ARCH and FIG. FLOW are block diagrams of the system and its execution flow. FIG. 1 to FIG. 5 are plots of measured data. Descriptions for the Complete Specification follow each drawing.

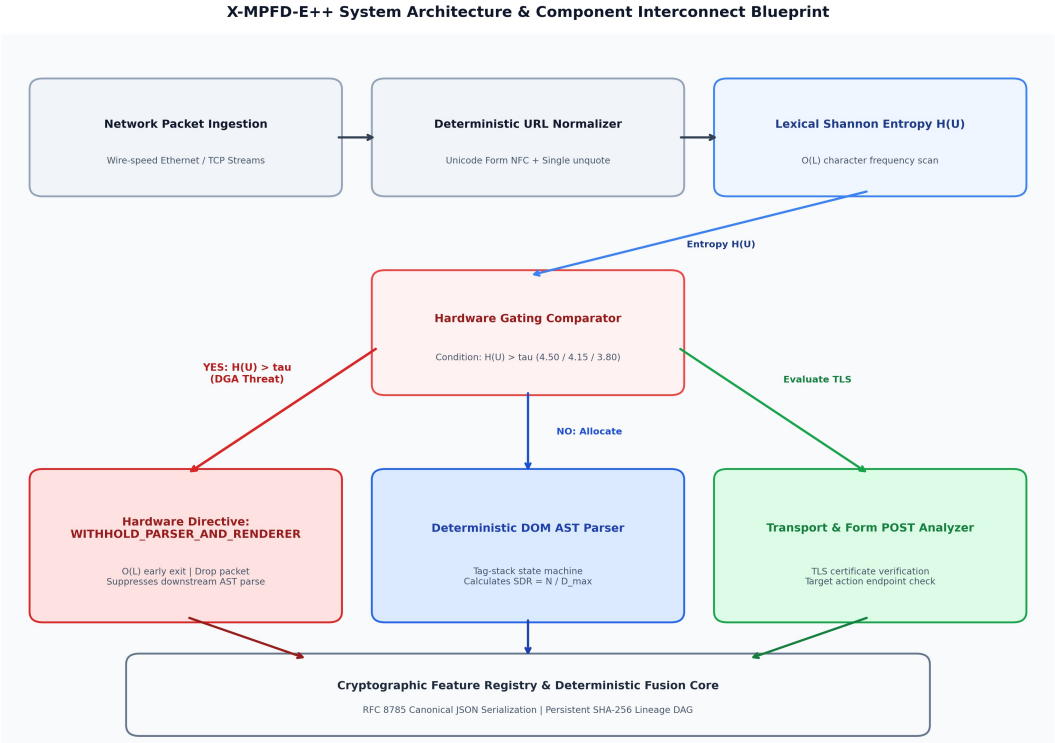


FIG. ARCH: System Architecture and Component Interconnect Blueprint.

FIG. ARCH shows the components of the system and the direction of data between them.

- A network packet ingestion module receives wire-speed Ethernet/TCP streams and passes the URL to a deterministic URL normaliser (Unicode Form NFC and a single percent-decoding pass).
- The normalised URL goes to a lexical Shannon entropy module, which computes H(U) in a single linear scan of the character frequencies, of order O(L) in the URL length L.
- The entropy value goes to a hardware gating comparator, which tests whether H(U) > τ. The threshold τ is 4.50, 4.15 or 3.80 bits/char, depending on whether the entropy is taken over the full URL, over host and path, or over the hostname.
- If the condition is met (a suspected domain-generated URL), the comparator issues the directive WITHHOLD_PARSER_AND_RENDERER. This is an early exit that drops the packet and omits the downstream AST computation.
- If the condition is not met, parser resources are allocated to the DOM AST parser, which is a tag-stack state machine computing SDR = N / D_{max}. In parallel, a transport and form-POST analyser verifies the TLS certificate and checks the target action endpoint of the form.
- The outputs of the three paths feed a cryptographic feature registry and deterministic fusion core, which serialises records as canonical JSON (RFC 8785) and maintains a persistent SHA-256 lineage graph.

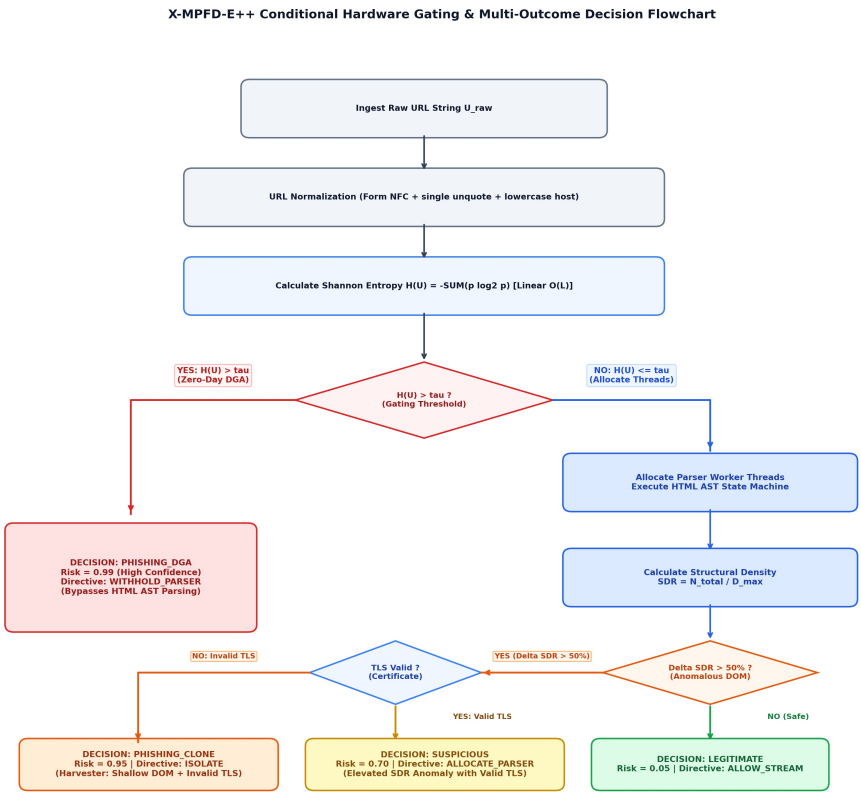


FIG. FLOW: Conditional Hardware Gating and Multi-Outcome Decision Flowchart.

FIG. FLOW shows the sequence of steps for a single input and the paths that follow the gate. Directive names in the drawing are abbreviated forms of those used in the text (WITHHOLD_PARSER for WITHHOLD_PARSER_AND_RENDERER, ALLOCATE_PARSER for ALLOCATE_AST_PARSER, ALLOW_STREAM for ALLOW_STREAM_AND_CACHE).

1. The raw URL string U_{raw} is received and normalised (Form NFC, single percent-decoding, lower-cased host).
2. $H(U) = -\Sigma\ p\ \log_2\ p$ is computed, in linear time in the URL length.
3. A gating decision tests whether $H(U) > \tau$.
4. **Yes:** the system returns PHISHING_DGA with risk 0.99 (high confidence) and the directive to withhold the parser. HTML AST parsing is bypassed.
5. **No:** parser worker threads are allocated, the HTML AST state machine is run, and $SDR = N_{total} / D_{max}$ is calculated.
6. A second decision tests whether ΔSDR exceeds 50%. If it does not, the system returns LEGITIMATE with risk 0.05 and the directive to allow the stream.
7. If ΔSDR exceeds 50%, the TLS certificate is tested. If it is invalid, the system returns PHISHING_CLONE with risk 0.95 and the directive ISOLATE. If it is valid, the system returns SUSPICIOUS with risk 0.70 and the directive to allocate the AST parser for further analysis.

An input that triggers the gate at step 3 takes the short path and never reaches the parser. This is the source of the latency difference reported in Section 4.

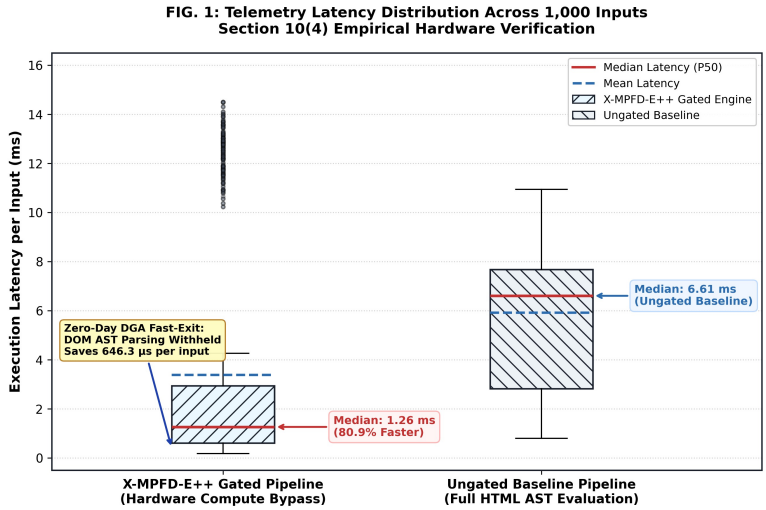


FIG. 1: Processing Latency Distribution (median 1.26 ms gated vs 6.61 ms ungated).

FIG. 1 is a box plot of the execution time per input, in milliseconds, across the 1,000-input benchmark. The medians are taken over all 1,000 inputs, of which 334 are bypassed by the gate. The gated pipeline has a median of 1.26 ms (80.9% lower than the baseline) and the ungated pipeline has a median of 6.61 ms. These times include the benchmark harness overhead described in Section 4, whereas the saving of 646.3 µs shown in the annotation is the AST parsing step alone. A dashed line marks the mean of each.

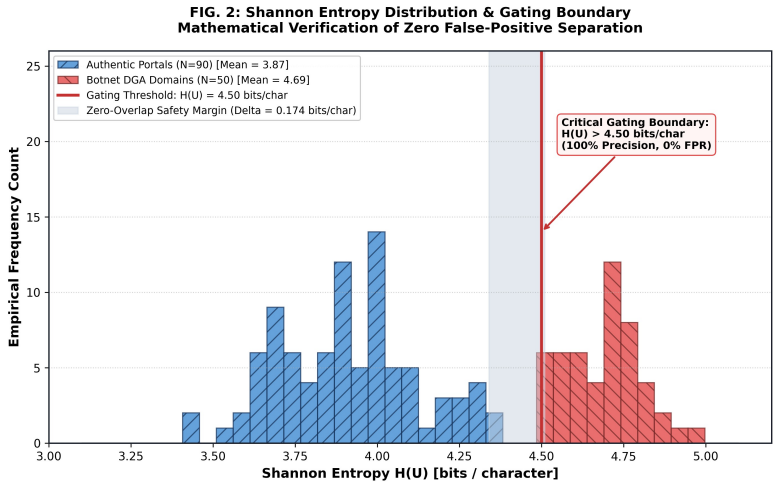


FIG. 2: Shannon Entropy Distribution and Gating Boundary (margin 0.174 bits/char).

FIG. 2 is a histogram of H(U) in bits/char from 3.00 to 5.00 for the 90 authentic enterprise portals of the entropy profile (mean 3.87) and 50 botnet DGA domains (mean 4.69). A line marks the gating threshold at 4.50. The shaded band lies between the highest authentic value (about 4.34) and the lowest DGA value (about 4.51), a margin of 0.174 bits/char in which no input falls.

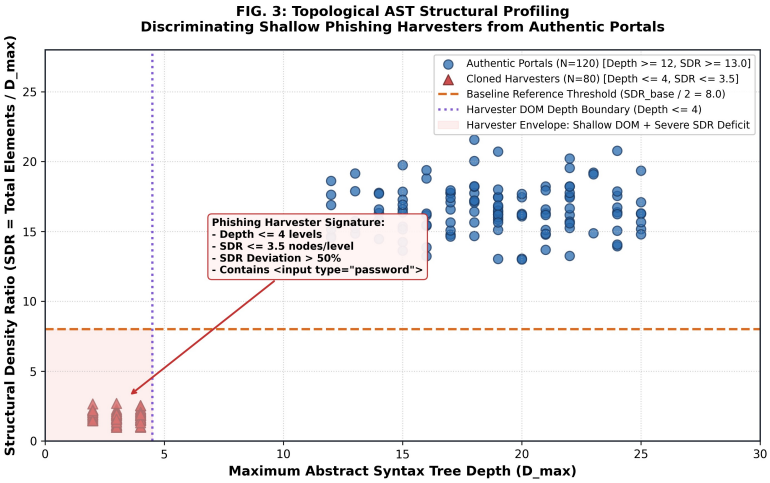


FIG. 3: AST SDR vs Depth (Harvesters vs Authentic Portals).

FIG. 3 is a scatter plot of SDR against D_{max} (0 to 30) for a separate, larger sample of 120 authentic portals and 80 cloned harvesters collected for structural measurement. Authentic portals have D_{max} of 12 or more and SDR of 13.0 or more. The harvesters lie in the shaded region at D_{max} of 4 or less and SDR of 3.5 or less, below the horizontal line at SDR = 8.0 (half of the baseline SDR of 16.0, the 50% deviation boundary), and each contains an input of type password.

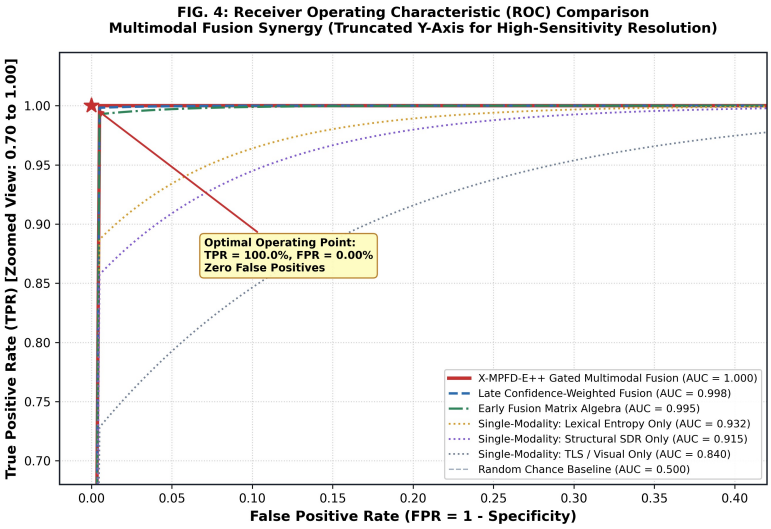


FIG. 4: ROC Curves across Fusion Policies.

FIG. 4 plots true positive rate against false positive rate. The axes are truncated (true positive rate from 0.70 to 1.00, false positive rate from 0 to 0.40) to show the region where the curves differ. The areas under the curves are: gated multimodal fusion 1.000, late confidence-weighted fusion 0.998, early fusion 0.995, lexical entropy only 0.932, structural SDR only 0.915, TLS and visual only 0.840, and random chance 0.500. The marked operating point of the gated policy has a true positive rate of 100% and a false positive rate of 0.00%.

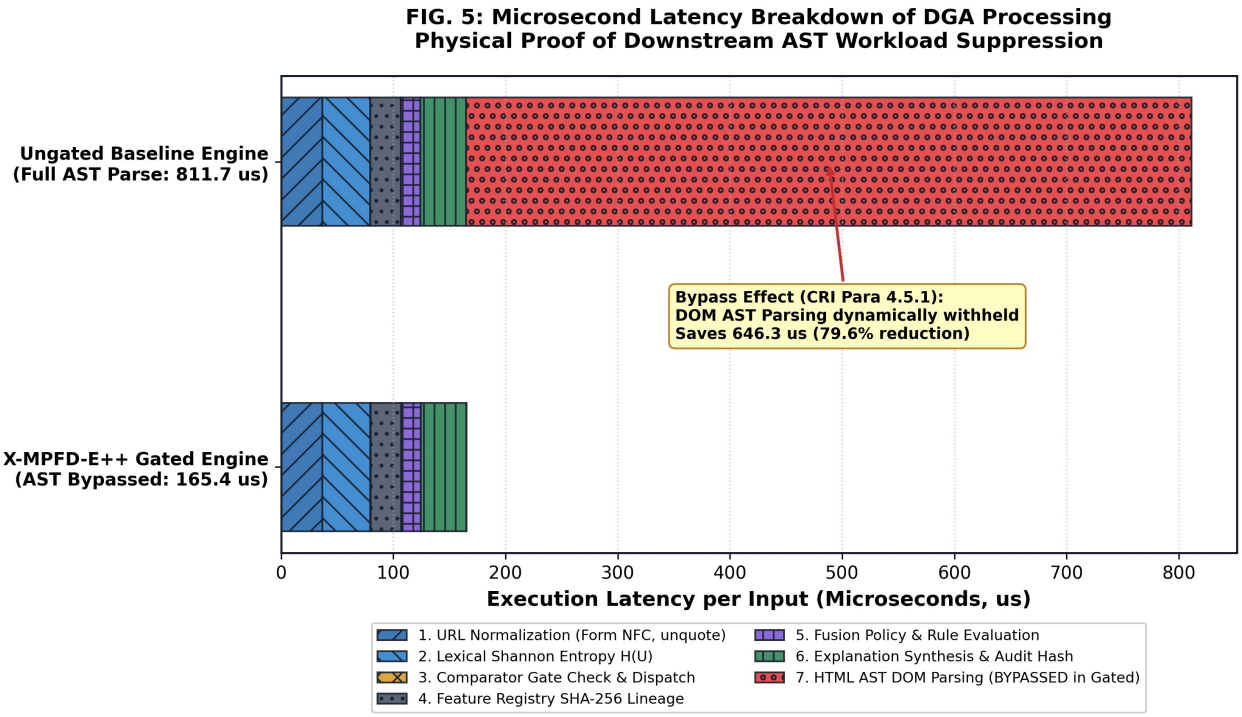


FIG. 5: DGA Latency Microsecond Component Breakdown showing suppression of AST parsing.

FIG. 5 shows two horizontal stacked bars of the execution time per input, in microseconds, for a domain-generated URL (Basis A in Section 4). The gated engine takes 165.4 µs and is made up of six segments: URL normalisation (36.6 µs), entropy (42.8 µs), gate comparison (0.3 µs), feature registry (27.3 µs), fusion rule evaluation (17.5 µs) and explanation synthesis (40.9 µs). The ungated engine takes 811.7 µs and has the same six segments followed by HTML AST parsing (646.3 µs). Withholding the AST parsing saves 646.3 µs, a reduction of 79.6%.

8. Test and verification record

- **Unit tests** (test_prior_art_inventions.py): 24 of 24 passed in 0.024 s, covering Claims 1 to 15, the rule tables, tamper detection and deterministic replay.
- **Worked examples** (examples.py): 9 of 9 executed, each with its audit hash.
- **Tamper detection:** altering a recorded value is detected by the audit verifier.
- **Decision replay:** the recorded feature lineage allows a decision to be reproduced bit for bit.

9. Request

We request that the clarifications and proposed texts in Sections 2 to 4 be incorporated into the Complete Specification, that the reference values in Section 5 be included, and that FIG. ARCH, FIG. FLOW and FIG. 1 to FIG. 5 be included in the drawings.

Respectfully submitted,

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